For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS **6032/1**

PAPER 1 Multiple Choice 1 hour

NOVEMBER 2025 SESSION

Additional materials:
Multiple Choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)
Scientific calculator

INSTRUCTIONS TO CANDIDATES

Answer **all** questions. For each question, there are **four** possible answers, **A**, **B**, **C** and **D**. Choose the correct answer. Record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

There are **forty** questions in this paper. Each correct answer will score **one** mark. Any rough working should be done on this question paper.

This question paper consists of 17 printed pages.

1 Which quantity is a vector?

A gravitational field strength
B magnetic flux
C gravitational potential
D internal energy

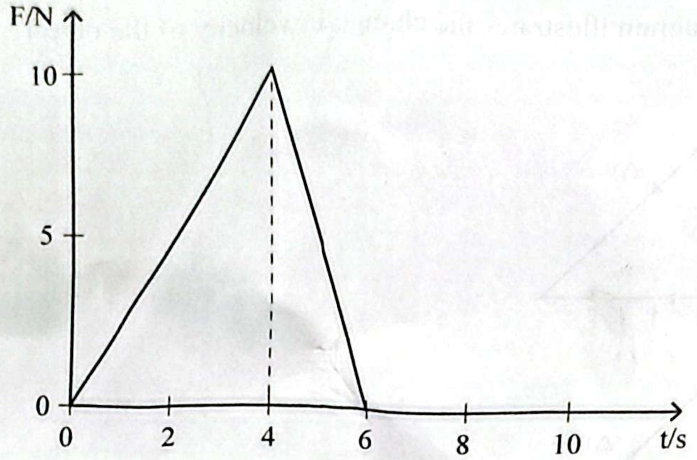
2 Which energy conversion is true for an object falling at terminal velocity?

A gravitational potential energy to kinetic energy
B kinetic energy to gravitational potential energy
C thermal energy to kinetic energy
D gravitational potential energy to thermal energy

3 Which statement is correct for an inelastic collision?

A total kinetic energy is conserved
B total force is conserved
C the velocity of approach is equal to velocity of separation
D total momentum is conserved

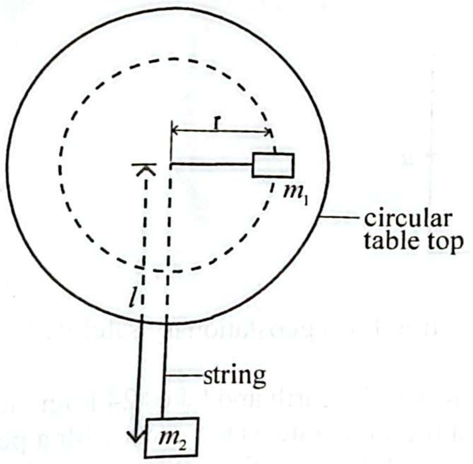
7 A 5.0 kg mass initially at rest is acted upon by a force, F, which varies with time, t, as shown in the diagram.



What is the velocity of the object at the end of 6 s?

A $0\ ms^{-1}$
B $5\ ms^{-1}$
C $6\ ms^{-1}$
D $30\ ms^{-1}$

8 The diagram shows a mass, m_1, on a frictionless table connected by a string passing through a hole at the centre of the table to a hanging mass m_2. Mass, m_1, is rotated with a speed, v, so that m_2 remains at rest.



What is the speed of rotation of m_1?

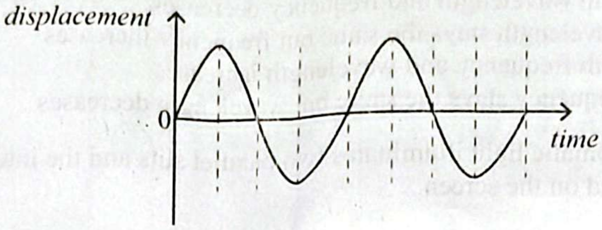
A $\frac{m_1 r}{m_2}\sqrt{\frac{g}{l}}$

B $\frac{m_2}{m_1}\sqrt{gr}$

C $\sqrt{\frac{m_2}{m_1}gr}$

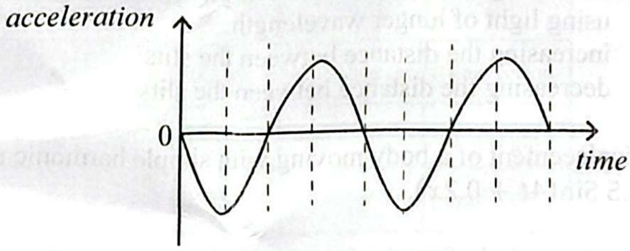
D $\sqrt{\frac{m_1}{m_2}gr}$

9 The graph shows the variation of *displacement* with *time* for a body performing simple harmonic motion.



Which graph is the corresponding *acceleration-time* graph?

A



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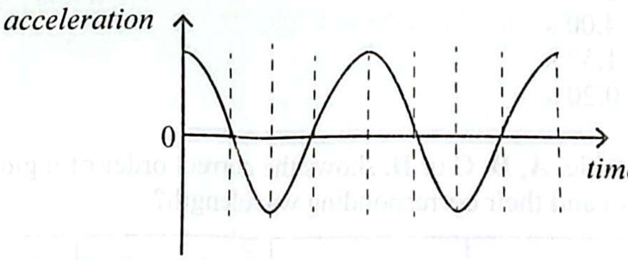

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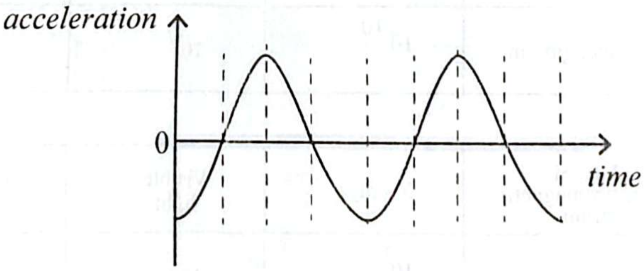
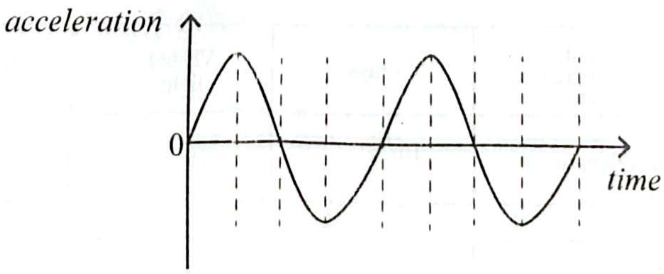


D



10 What happens when monochromatic light moves from air to glass?

A both wavelength and frequency decreases
B wavelength stays the same but frequency increases
C both frequency and wavelength increases
D frequency stays the same but wavelength decreases

11 Monochromatic light illuminates two parallel slits and the interference pattern which results is observed on the screen.

Which one will reduce the fringe separation?

A increasing the distance between the slits and the screen
B using light of longer wavelength
C increasing the distance between the slits
D decreasing the distance between the slits

12 The displacement of a body moving with simple harmonic motion is given by the equation $y = 3.5\,\mathrm{Sin}(4t + 0.2x)$.

What is the period of motion?

A 3.52 s
B 4.00 s
C 1.57 s
D 0.20 s

13 Which table, **A**, **B**, **C** or **D**, shows the correct order of regions of the electromagnetic spectrum and their corresponding wavelength?

A

regions of electromagnetic spectrum	X - rays	Visible light	Infrared radiation	Radio waves
wavelength / m	10^{-10}	10^{-7}	10^{-5}	10^{3}

B

regions of electromagnetic spectrum	X - rays	Visible light	Infrared radiation	Radio waves
wavelength / m	10^{3}	10^{-5}	10^{-7}	10^{-10}

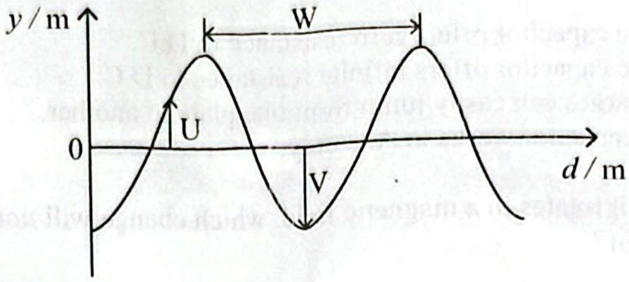
C

regions of electromagnetic spectrum	X - rays	Visible light	Infrared radiation	Radio waves
wavelength / m	10^{-10}	10^{-7}	10^{3}	10^{-5}

D

regions of electromagnetic spectrum	X - rays	Visible light	Infrared radiation	Radio waves
wavelength / m	10^{-7}	10^{-10}	10^{-5}	10^{3}

14 The diagram shows a progressive wave profile.



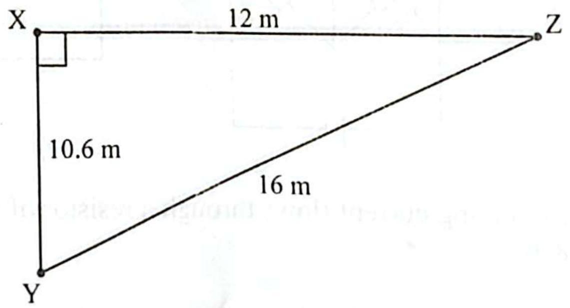
Which list correctly identifies U, V and W?

	U	V	W
A	amplitude	displacement	time period
B	displacement	amplitude	wavelength
C	displacement	time period	amplitude
D	amplitude	displacement	wavelength

15 What is the maximum order number of visible light with a wavelength of 600 nm when it is passed through a diffraction grating with 450 lines per millimetre?

A 3
B 4
C 6
D 7

16 The diagram show distances travelled by two waves of wavelength 4 m. They travel from X and Y to Z.



What is the phase difference of the waves at X and Y, which makes them to reach point Z in phase?

A $\frac{2\pi}{12}$ radians

B $\frac{2}{8}\pi$ radians

C π radians

D 2π radians

[Turn over

17 Which statement is true for a capacitor?

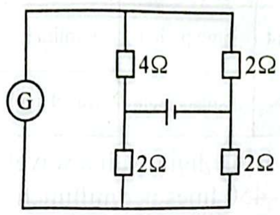
A The capacitor offers zero resistance to D.C.
B The capacitor offers infinite resistance to D.C.
C Charges can easily jump from one plate to another.
D A capacitor works in A.C only.

18 When a coil rotates in a magnetic field, which change will **not** affect the magnitude of the induced emf?

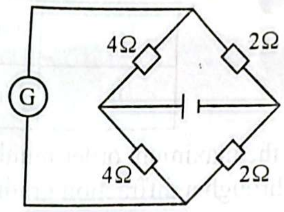
A changing the resistance of the coil
B changing the magnetic flux density
C changing the speed of the coil
D changing the number of turns in the coil

19 In which circuit will the centre zero galvanometer show a deflection?

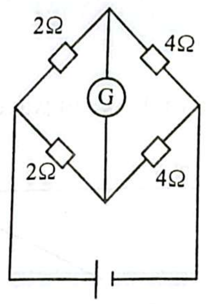
A



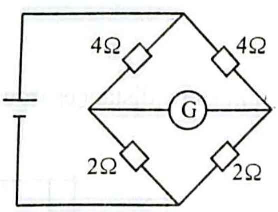
B



C



D



20 When an alternating current flows through a resistor of resistance 12 Ω, heat is dissipated at a rate of 48 W.

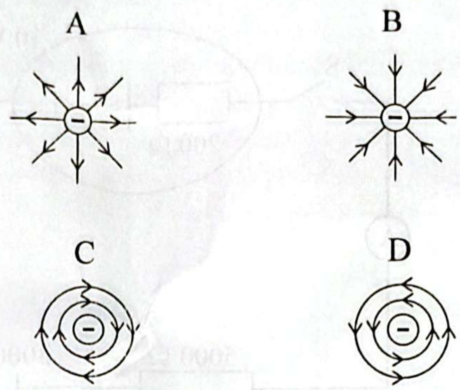
What is the peak value of the alternating current?

$P = I^2 R$

$I = \sqrt{\frac{P}{R}}$

A 4.00 A
B 2.83 A
C 2.00 A
D 1.41 A

21 Which diagram shows the electric field pattern for an isolated negative point charge?



22 Logic gates X and Y have the truth tables shown.

Truth table for X

inputs		output
P	Q	R
0	0	1
0	1	1
1	0	1
1	1	0

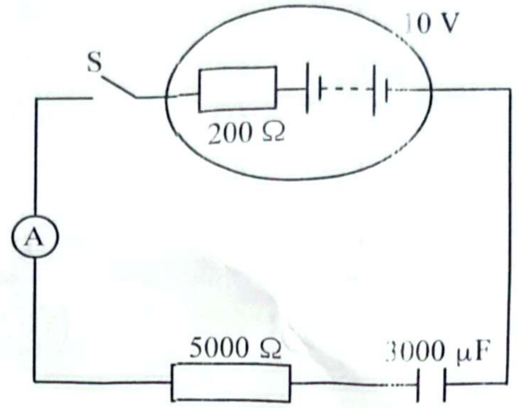
Truth table for Y

input	output
0	1
1	0

Which single logic gate is formed when the output of X is connected to the input of Y?

A NAND
B AND
C NOR
D NOT

23 The circuit shows a battery of emf 10 V and internal resistance 200 Ω connected in series with a resistor of resistance 5 000 Ω and an uncharged capacitor of capacitance 3 000 μF.



When switch S is closed, the ammeter reads 0.80 mA at a particular time.

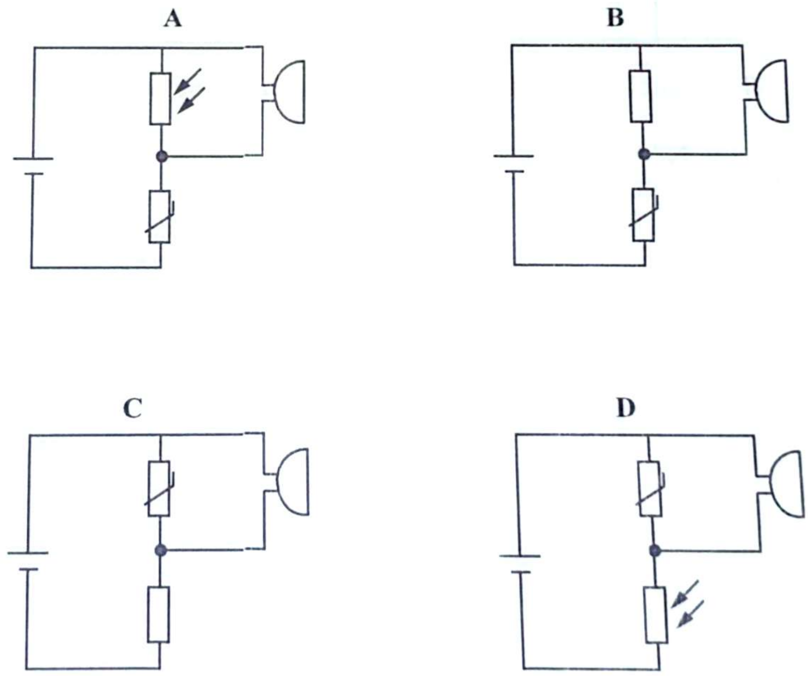
What is the potential difference across the capacitor when the ammeter reads 0.80 mA?

TR + r

A 4.00 V
B 4.16 V
C 5.84 V
D 6.00 V

24 In the circuits, **A**, **B**, **C** and **D**, shown, temperature remains constant. The buzzer produces sound when potential difference across it increases.

In which circuit, **A**, **B**, **C** or **D**, does increasing light intensity cause the buzzer to produce sound?



25 The equation for the Young's Modulus, E, of a material is $E = \frac{Fl}{Ae}$. The length of the material is l, extension, e and force, F.

Which equation shows the effect of halving the cross-sectional area, A, on the Young's Modulus?

A $E = \frac{Ae}{2Fl}$

B $E = \frac{Fl}{2Ae}$

C $E = \frac{2Fl}{Ae}$

D $E = \frac{2Ae}{Fl}$

26 A substance has a temperature of θ °C.

What would be the temperature of the substance on the thermodynamic temperature scale?

A $(273 - \theta)$ K
B $(273 + \theta)$ K
C $(100 - \theta)$ K
D $(100 + \theta)$ K

27 A constant force, F, was applied onto a material and an extension, x was produced. The Young's Modulus of the material is E.

What is the work done in stretching the material?

A $\frac{1}{2}Fx$

B $\frac{1}{2}Ex$

C Fx

D $\frac{Fx}{E}$

28 What is the change in internal energy when 20 kJ of work is done on the system and 30 kJ of heat is given out by the system?

A +40 kJ
B −40 kJ
C +10 kJ
D −10 kJ

[Turn over

29 A metal ball bearing of mass, m, and specific heat capacity, c, moving with speed, v, is brought to rest. All its kinetic energy is converted to thermal energy which it absorbs causing a temperature rise ΔT.

What is the value of v?

A $\frac{1}{2}c\,\Delta T$

B $2c\,\Delta T$

C $\sqrt{c\,\Delta T}$

D $\sqrt{2c\,\Delta T}$

30 Which pair is correct for a constant volume gas thermometer?

	advantage	disadvantage
A	very sensitive	bulky
B	very accurate	not sensitive
C	wide range	not accurate
D	measures rapidly changing temperature	slow response

31 What is the effect of reducing the temperature by half on the root mean square speed, *c*, of gas molecules?

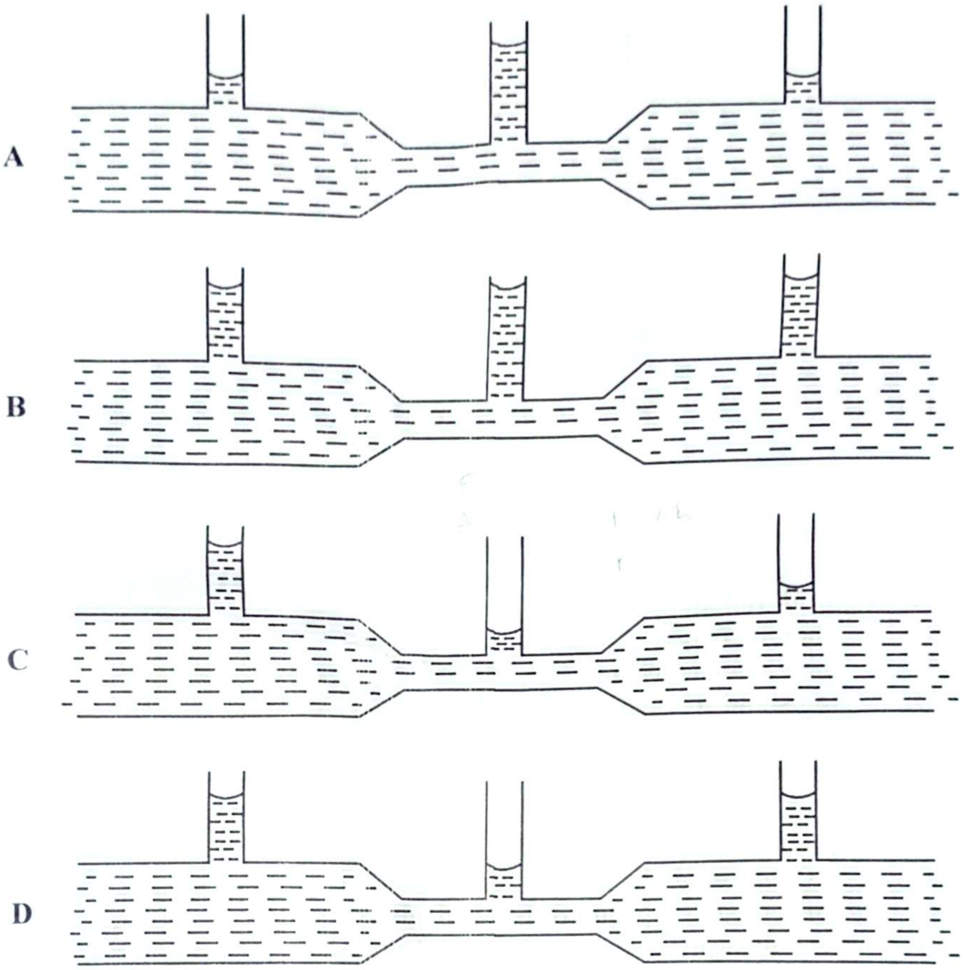
A $\sqrt{\frac{c}{2}}$

B $\frac{c}{\sqrt{2}}$

C $2\,c$

D $\sqrt{2}\,c$

32 Which diagram shows the correct height reached by a fluid for a constricted horizontal tube?



33 An oil drop carrying a charge, e, remains stationary between two horizontal metal plates which are separated by a distance, x, and maintained at a potential of V. If g is the acceleration of freefall, the mass of the oil drop is

A $\frac{ex}{Vg}$.

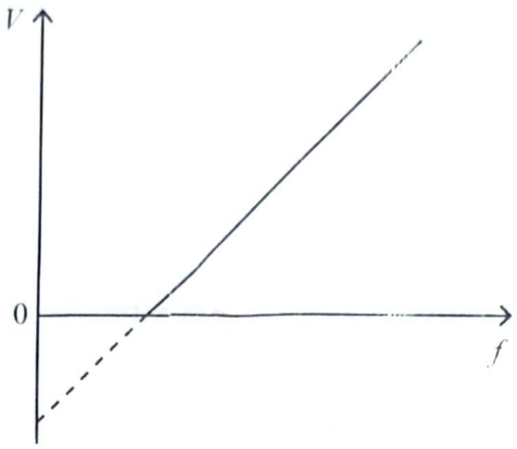
B $\frac{eV}{xg}$.

C $\frac{eVx}{g}$.

D $\frac{V}{xg}$.

[Turn over

34 The graph shows variation of stopping potential, V, with frequency, f, of incident radiation.



What does the intercept on the frequency, f axis represent?

A work function
B photon energy
C the Planck constant
D threshold frequency

35 Which phenomenon suggests the wave-like nature of electrons?

A electron transitions between energy levels
B diffraction of electrons by thin metal foil
C line emission spectra of isolated atoms
D line absorption spectra of isolated atoms

36 An electron absorbs a photon and makes a transition from a level of energy – 3.40 eV to one of energy – 0.85 eV.

What is the energy of the photon?

A – 4.25 eV
B – 2.55 eV
C 2.55 eV
D 4.25 eV

37 Which statement is true about nuclear fusion?

A the products are more stable than the reactants
B the products are less stable than the reactants
C total mass of products is equal to total mass of reactants
D total mass of products is greater than total mass of reactants

38 The following are statements about the alpha-particle scattering experiment.

P: the nucleus contains protons
Q: the bulk of atomic space is empty
R: the nucleus has the bulk of atomic mass

Which statements are conclusions from the experiment?

A P and R
B P and Q
C Q and R
D P, Q and R

39 What is the energy equivalent of a mass defect of 8.68×10^{-28} kg?

A 7.81×10^{-11} J

B 3.91×10^{-11} J

C 2.60×10^{-19} J

D 6.78×10^{-30} J

40 A carrier wave of frequency 150 kHz is amplitude modulated by a signal of frequency 10 kHz.

What are the two sideband frequencies?

A 10 kHz and 150 kHz
B 15 kHz and 1 500 kHz
C 140 kHz and 160 kHz
D 145 kHz and 155 kHz